HMC Sanaad B2E Platform

Project Handover & Developer Onboarding

HMC_BackEnd (Sanaad API) · HMC_Gateway (Public Gateway)

NestJS 11 · Node.js 20 · Oracle · SQL Server

Version 1.0 — 2026-08-30

Internal document — contains no credentials.

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Version: 1.0 · **Date:** 2026-08-30 · **Projects covered:** `HMC_BackEnd` (Sanaad API) + `HMC_Gateway` (public gateway)

Audience: a developer who is new to this project and has little or no experience with Node.js, TypeScript, or NestJS.

Goal: after reading this document you should be able to run, understand, debug, maintain, and extend the system with minimal help from the previous developer.

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1. Project Overview

Project in 5 Minutes

Sanaad is HMC's (Hamad Medical Corporation) employee self-service mobile app. Employees use it to see their payslips, request leave, update their personal details, request HR letters, manage dependents, book staff-clinic appointments, approve requests from their team, and more.

All the *real* business logic and data already exist in HMC's **Oracle HR system** (as database views and PL/SQL procedures named `XXHMC_SND_*`) and in a legacy **SQL Server** database (login devices, MPINs, OTPs). This project does **not** re-implement that logic. Instead, it is a modern, secure **API layer** that:

1. Lets the mobile app log in safely (OTP by SMS + a 4-6 digit MPIN → a JWT token),
2. Translates clean REST API calls (e.g. `GET /api/v1/leave/balance`) into the correct Oracle view reads and procedure calls,
3. Returns results in the exact JSON shape the mobile app expects (including Arabic/English localization).

There are **two applications**:

```
Mobile App ──HTTPS──▶ HMC_Gateway ──HTTP(S)──▶ HMC_BackEnd ──▶ Oracle HR (business data)
                      (public door)            (the API)   ──▶ SQL Server "Users DB" (devices, MPIN)
                                                           ──▶ SQL Server "MOTC_SMS" (OTP via SMS outbox)
                                                           ──▶ Cerner (clinic appointments, HTTP)
                                                           ──▶ LDAP / Entra ID (employee directory)
```

- **HMC_Gateway** is the only thing exposed to the internet. It forwards login-journey requests to the backend, checks the JWT on every other request (without calling the backend), applies rate limiting against brute-force attacks, and proxies everything else through untouched.
- **HMC_BackEnd** is the actual API: 14 feature modules re-exposing the 71 legacy "Sanaad operations".

Main users: HMC employees (mobile app) — plus developers/testers who use the built-in diagnostics tools (`/docs` Swagger UI, `/dev-console`, `/api-logs`, `/diagnostics/*`).

Key design idea to remember: *the database is the source of truth for business logic; this codebase is a disciplined translator.* Most bugs are solved by looking at what Oracle/SQL Server actually expects — the project has excellent built-in tooling for exactly that (see Section 14).

Main Features (by module)

Area	What the employee can do
Auth	Log in with OTP + MPIN, refresh session, logout, forgot MPIN
Profile / Employee	View & update personal details, view employment/salary history, change supervisor
Payslip	List pay periods, generate a full payslip
Leave	Check balance, apply/amend/cancel leave, return from leave, leave history
Letters	Request HR letters/certificates
Identity	View/update QID, request company ID card
Contact	Update phones and addresses
Dependents	Add/update/delete dependents, passport details
School Fees	Apply for children's school-fee support
Appointments	Book staff-clinic appointments (via Cerner)
Annual Ticket	Apply for / cancel the annual flight ticket benefit
Approvals	Managers approve/reject/reassign requests, worklist

High-Level Architecture

Every backend feature module follows the same **Clean Architecture** layering:

```
interface/       ← Controller + DTOs (HTTP shape, validation, Swagger docs)
application/     ← Service (orchestration, no SQL here)
domain/          ← Repository interface ("port") + domain types — framework-free
infrastructure/  ← Oracle/SQL/HTTP adapter implementing the port
```

Cross-cutting concerns (config, database pools, JWT auth, logging, error handling, response envelope) live once in `src/core/` and apply to every request automatically.

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2. Technology Stack

Technology	Version	Purpose	Used In
Node.js	20.x (Docker uses `node:20-slim`)	JavaScript runtime	Both projects
TypeScript	5.x	Typed language, compiled to JS	Both projects
NestJS	^11.1.17	Application framework (modules, DI, HTTP)	Both projects
Express	(via `@nestjs/platform-express`)	Underlying HTTP server	Both projects
node-oracledb	^6.5.0	Oracle database driver (connection pool)	Backend
mssql (tedious)	^11.0.1	SQL Server driver — two pools (Users DB, MOTC SMS DB)	Backend
@nestjs/jwt + passport-jwt	^11.0.2 / ^4.0.1	Issue & verify JWT tokens	Both (backend issues, both verify)
ldapts	^9.0.0	LDAPS directory lookups	Backend
@nestjs/axios + axios	^4.0.1 / ^1.6.8	Outbound HTTP (Cerner, Entra Graph, SMS) & proxying	Both
@nestjs/config + Joi	^4.0.4 / ^17.13.0	Typed configuration + env validation at boot	Both
class-validator / class-transformer	^0.14.1 / ^0.5.1	Request DTO validation	Both
@nestjs/swagger	^11.2.0	OpenAPI docs at `/docs`	Both
@nestjs/throttler	^6.4.0	Rate limiting on login endpoints	Gateway
helmet / compression	^8.1.0 / ^1.7.4	Security headers / gzip	Both
Jest + ts-jest + Supertest	^29.x / ^7.0.0	Unit & e2e testing	Both
Docker (multi-stage)	node:20-slim + Oracle Instant Client	Container deployment	Both

> Note: `HMC_BackEnd/README.md` still says "NestJS 10" — it is outdated; `package.json` is the truth (v11).

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3. Environment Setup

3.1 Prerequisites

1. **Node.js 20 LTS** — install from https://nodejs.org (check with `node -v`).
2. **Git**.
3. **Visual Studio Code** (recommended) with the ESLint and Prettier extensions.
4. Optional for full local runs: access to the HMC network (Oracle/SQL Server/LDAP are internal). **You can run everything without any database** — see 3.5.

> **Windows note (important):** on the current dev machines, PowerShell's execution policy blocks the `npm` command wrapper. Always use **`npm.cmd`** and **`npx.cmd`** instead of `npm` / `npx`.

3.2 Get the code and install

```
git clone <repository-url>            # ← ask the team for the actual repo URL(s)
cd development/HMC_BackEnd
npm.cmd ci                            # install exact locked dependencies

cd ../HMC_Gateway
npm.cmd ci
```

`npm.cmd ci` reads `package-lock.json` and installs the exact dependency tree (preferred over `npm install` for reproducibility).

3.3 Configure environment variables

Each project reads a `.env` file at its root. Start from the documented template:

```
# in HMC_BackEnd/
copy .env.example .env
# in HMC_Gateway/
copy .env.example .env
```

Open each `.env` and fill in what you have. Every variable is explained inside `.env.example` and in Section 12. **Never commit `.env`.**

3.4 Build and start

```
# Backend (from HMC_BackEnd/)
npm.cmd run build          # compile TypeScript → dist/
npm.cmd run start:dev      # OR: hot-reload development server

# Gateway (from HMC_Gateway/)
npm.cmd run build
npm.cmd run start:dev
```

3.5 Run WITHOUT any databases (the standard local smoke test)

The backend boots happily with every external system disabled — dependency-injection still wires up, all routes register, and Swagger works. This is how you verify your environment quickly:

```
# Terminal 1 — HMC_BackEnd
cd HMC_BackEnd
$env:ORACLE_DISABLED='true'; $env:AUTH_DISABLED='true'; $env:LDAP_ENABLED='false'
$env:JWT_SECRET='local_dev_secret_local_dev_secret_1234'; $env:PORT='3009'
node dist/main.js

# Terminal 2 — HMC_Gateway
cd HMC_Gateway
$env:BACKEND_BASE_URL='http://localhost:3009'; $env:BACKEND_API_PREFIX='api/v1'
$env:JWT_SECRET='local_dev_secret_local_dev_secret_1234'; $env:AUTH_DISABLED='true'
$env:GATEWAY_PORT='3001'
node dist/main.js
```

3.6 Verify it works

Check	URL	Expect
Backend alive	`http://localhost:3009/api/v1/health`	`{"status":"ok", ...}` with oracle/usersDb/motcSmsDb blocks
Backend API docs	`http://localhost:3009/docs`	Swagger UI listing ~106 endpoints
Gateway alive	`http://localhost:3001/api/v1/health`	`{"status":"ok"}`
Gateway → backend	`http://localhost:3001/api/v1/health/backend`	backend's health relayed
Full proxy path	`POST http://localhost:3001/api/v1/auth/login` (any JSON body with `username`, `mpin`, `imeinumber`, `platform`, `version`)	a signed JWT (dev bypass)

3.7 Docker (optional)

Both projects ship a multi-stage `Dockerfile` (backend includes the Oracle Instant Client) and a `docker-compose.yml`:

```
cd HMC_BackEnd
docker compose up --build
```

Compose loads `.env` via `env_file` and additionally maps a specific list of host variables (Oracle/LDAP/Users-DB/SMS) in its `environment:` block — host values win over `.env`. ⚠️ Newer variables (`MOTC_SMS_*`, `OTP_STORE`, `DIAGNOSTICS_ENABLED`, `AUTH_STATIC_LOGIN`) are **not** in that block yet, so in Docker they must be set in `.env` (see Section 21).

> DNS inside containers: short hostnames like `HPDSLDEVV01` do **not** resolve inside Linux containers. Use the FQDN (e.g. `HPDSLDEVV01.hmc.org.qa`) or an IP, or add an `extra_hosts` entry.

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4. Required Node.js + TypeScript Basics

Only what you need for *this* project.

npm & package.json

- **What:** npm is Node's package manager; `package.json` declares dependencies and runnable scripts.
- **Why:** everything you run goes through it (`npm.cmd run build` , `npm.cmd test`).
- **Here:** each project has its own `package.json` . The `scripts` section is your command menu (Section 17). `package-lock.json` pins exact versions — commit it, don't edit it by hand.

async/await & Promises

- **What:** a `Promise` represents a value that arrives later (e.g. a database answer). `await` pauses the function until it arrives; such functions are marked `async` .
- **Why:** every database/HTTP call in this codebase is asynchronous.
- **Here:** e.g. `HMC_BackEnd/src/modules/leave/infrastructure/oracle/leave.oracle.repository.ts` — `async getBalance(...)` awaits an Oracle call. Fan-out reads use `Promise.all([...])` to run several reads in parallel (e.g. `profile.oracle.repository.ts`).

Environment variables

- **What:** key=value settings read from the process environment (or a `.env` file) at startup.
- **Why:** the same code runs in dev/staging/production with different databases and secrets.
- **Here:** loaded by `@nestjs/config` from `.env` , validated by Joi (`src/core/config/env.validation.ts`), and exposed as typed objects (`src/core/config/configuration.ts`). Code never reads `process.env` directly outside that file.

TypeScript basics

- **What:** JavaScript + types. Compiled to plain JS in `dist/` by `npm.cmd run build` .
- **Why:** types catch mistakes at build time; the whole codebase relies on them.
- **Key concepts used here:**
  - **Interfaces** — describe object shapes, e.g. `LeaveBalanceQuery` in `src/modules/leave/domain/leave.repository.ts` ; repository "ports" are interfaces (`OtpPort` , `LovRepository`).
  - **Classes** — controllers/services/DTOs are classes because NestJS decorators attach to them.
  - **Decorators** — `@Controller()` , `@Get()` , `@Injectable()` , `@IsString()` ... metadata annotations NestJS/class-validator read at runtime.
  - **Generics** — `query<T>(...): Promise<T[]>` lets a helper return typed rows.
  - **Path aliases** — imports like `@core/...` , `@shared/...` map to `src/core` , `src/shared` (see `tsconfig.json` → `paths`).
- **Modules (ES/TS)** — every file exports things (`export class ...`) and imports them; not to be confused with *NestJS modules* (next section).

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5. NestJS Basics for This Project

NestJS organizes an app into **modules** containing **controllers** (HTTP layer) and **providers/services** (logic), glued together by **dependency injection**.

Modules (`@Module`)

A module groups related pieces and declares what it imports/provides/exports.

- Root: `src/app.module.ts` imports `CoreModule` + all feature modules.
- Example: `src/modules/leave/leave.module.ts` declares the leave controllers, service, and binds the repository interface to its Oracle implementation.

Controllers (`@Controller` , `@Get` , `@Post`)

Receive HTTP requests, validate input via DTOs, call a service, return the result.

- Example: `src/modules/leave/interface/leave.controller.ts` — `@Controller('leave')` + `@Get('balance')` handles `GET /api/v1/leave/balance` .
- Controllers contain **no business logic and no SQL**.

Services / Providers (`@Injectable`)

Hold orchestration logic; injected into controllers via constructors.

- Example: `src/modules/leave/application/leave.service.ts` .

Dependency Injection (DI)

You never write `new LeaveService()` . You declare constructor parameters and Nest supplies instances. Interfaces are bound via **tokens**:

```
// leave.module.ts
{ provide: LEAVE_REPOSITORY, useClass: LeaveOracleRepository }
// leave.service.ts
constructor(@Inject(LEAVE_REPOSITORY) private readonly repo: LeaveRepository) {}
```

This is why swapping an implementation (e.g. OTP store `motc` vs `legacy` in `src/modules/auth/auth.module.ts`) is a one-line factory, not a rewrite.

DTOs + ValidationPipe

DTO = a class describing a request's shape with validation decorators:

```
export class LeaveBalanceQueryDto extends LangQueryDto {
  @IsString() @IsNotEmpty() username?: string;
}
```

A **global ValidationPipe** (`src/core/core.module.ts`) runs with `whitelist: true, forbidNonWhitelisted: true` — unknown JSON keys are **rejected with 400**, and declared fields are validated/transformed automatically.

Guards

Run before the handler and decide "may this request proceed?".

- `JwtAuthGuard` (global) — requires a valid bearer token unless the route is `@Public()` .
- `RolesGuard` — enforces `@Roles(...)` .
- `FunctionAccessGuard` — enforces `@RequireFunction('CODE')` against the `functions` claim in the JWT.
- `DiagnosticsEnabledGuard` — hides diagnostics routes (404) when `DIAGNOSTICS_ENABLED=false` . All in `src/core/auth/` and `src/core/http/` .

Middleware

`CorrelationIdMiddleware` (`src/core/http/correlation-id.middleware.ts`) gives every request an `x-correlation-id` (kept in `AsyncLocalStorage`) so one request's log lines can be tied together across layers.

Interceptors

Wrap around the handler (before + after). Registered globally in order: `ApiLogInterceptor` (records every request/response) → `AuditInterceptor` (audit trail) → `TimeoutInterceptor` (408 after `REQUEST_TIMEOUT_MS`) → `ResponseInterceptor` (wraps results in the Sanaad envelope + Arabic/English localization;

`@SkipEnvelope()` opts out — the auth journey uses it).

Exception Filters

`AllExceptionsFilter` (`src/core/http/all-exceptions.filter.ts`) catches **every** error, classifies it (validation / DB / external service / auth…), logs the full details server-side, and returns a safe, consistent JSON error to the client.

Decorators cheat-sheet (project-specific)

Decorator	Meaning	Defined in
`@Public()`	Skip JWT guard	`core/auth/decorators/public.decorator.ts`
`@CurrentUser()`	Inject the authenticated user	`core/auth/decorators/current-user.decorator.ts`
`@Roles(...)` / `@RequireFunction('code')`	Authorization	`core/auth/decorators/`
`@SkipEnvelope()`	Return raw JSON (no Sanaad envelope)	`core/http/response.interceptor.ts`
`@ApiReadOkResponse/@ApiActionOkResponse`	Swagger response docs	`shared/swagger/`
`@VerifiedBody(...)`	Swagger body with a real, staging-verified example	`shared/dto/verified-body.ts`

---

6. Complete Project Structure

6.1 Workspace

```
development/
├── AGENTS.md                ← ⭐ operational knowledge base: verified DB facts, build quirks, staging findings. READ IT.
├── Docs Project/            ← client-provided legacy service mapping (sanaad-api-service-mapping.html = proc-parameter truth)
├── Docs_Ai/                 ← architecture & domain documentation referenced by the README
├── PROJECT_ANALYSIS.md      ← the technical analysis behind this document
├── HMC_BackEnd/             ← the Sanaad API
└── HMC_Gateway/             ← the public gateway
```

6.2 HMC_BackEnd/src

```
src/
├── main.ts                  ← bootstrap: 15mb body limit, helmet, compression, CORS, /api/v1 prefix, Swagger (/docs)
├── app.module.ts            ← imports CoreModule, LookupsModule + 13 feature modules
│
├── core/                    ← cross-cutting, applies to every request
│   ├── core.module.ts       ← global pipe/interceptors/guards/filter registration (ORDER MATTERS — see §13)
│   ├── config/              ← configuration.ts (15 typed namespaces) + env.validation.ts (Joi, fails boot on bad env)
│   ├── database/
│   │   ├── oracle.service.ts          ← THE Oracle pool: query/call/callCursor/callMultiCursor + per-call logging
│   │   ├── oracle-schema.service.ts   ← asks Oracle's data dictionary for real column/parameter names (never guess!)
│   │   ├── base.repository.ts         ← base class every Oracle adapter extends (readByResolvedKey, callSubmitProc, callRowsProc…)
│   │   ├── mssql.service.ts           ← SQL Server pool #1: Users/Sanaad DB (devices, MPIN, legacy OTP, app tables)
│   │   ├── motc-sms-db.service.ts     ← SQL Server pool #2: MOTC_SMS DB (OTP push table = SMS delivery)
│   │   ├── diagnostics.controller.ts  ← /diagnostics/*: oracle-object describe, oracle call logs, SQL consoles
│   │   ├── oracle-log.store.ts        ← in-memory record of EVERY Oracle call (sql, binds, duration, ORA code)
│   │   └── sql-console.util.ts        ← SELECT-only enforcement for the SQL consoles
│   ├── auth/                ← JwtStrategy, JwtAuthGuard, RolesGuard, FunctionAccessGuard, TokenRevocationService, decorators
│   ├── http/                ← ResponseInterceptor (envelope+localization), TimeoutInterceptor, AllExceptionsFilter,
│   │                          CorrelationIdMiddleware, DiagnosticsEnabledGuard, exception classifier
│   ├── health/              ← /health (+ /health/db, /health/users-db, /health/motc-sms-db connectivity probes)
│   ├── audit/               ← 5-level audit trail → JSON log lines
│   ├── logging/             ← ApiLogInterceptor + /api-logs REST + HTML dashboard (+file writer, redaction)
│   └── dev-console/         ← /dev-console: SQL worksheet, Oracle object browser, API tester (dev/staging only)
│
├── shared/                  ← reusable, dependency-free building blocks
│   ├── constants/oracle-objects.ts    ← ⭐ allow-list of ~95 XXHMC_SND_* names — ONLY these can be queried
│   ├── constants/oracle-columns.ts    ← key-column candidate lists for view filtering
│   ├── constants/lov-names.ts         ← public lovname/lookupname → Oracle object registry
│   ├── constants/error-codes.ts       ← ORA-code extraction & HTTP mapping
│   ├── domain/  (lang.ts, lov-item.ts, submit-result.ts)
│   ├── dto/     (lang-query, common-query, oracle-submit helpers, pagination, envelopes)
│   ├── utils/   (date.util, localize.util ⭐ AR/EN twin collapsing, mapper.util, blob.util, url-decode.util)
│   └── swagger/ (response decorators, attachment body helpers)
│
├── lookups/                 ← shared-kernel LOV service (cached Oracle view reads) — used by ALL feature modules
│   ├── interface/lookups.controller.ts   (/lookups/yes-no, /rfmi-user, /lov, /master)
│   ├── application/lookups.service.ts
│   └── infrastructure/oracle/lov.oracle.repository.ts  (cache, request coalescing, dictionary-resolved filters)
│
└── modules/                 ← 13 feature modules, all with the same internal layout:
    ├── auth/                ←  interface/ (controller+DTOs) · application/ (services) · domain/ (ports) · infrastructure/ (adapters)
    ├── profile/  employee/  payslip/  leave/  letters/  identity/
    ├── contact/  dependents/ school-fees/ appointments/ annual-ticket/ approvals/
```

6.3 HMC_Gateway/src

```
src/
├── main.ts                 ← helmet, compression, CORS, /api/v1 prefix, Swagger (gateway-owned routes only)
├── app.module.ts           ← CoreModule → AuthModule → ProxyModule (wildcard imported LAST)
├── core/
│   ├── core.module.ts      ← config+Joi, ThrottlerModule, global ValidationPipe + JwtAuthGuard + AllExceptionsFilter
│   ├── auth/               ← local JWT validation (shared secret with backend) + @Public/@CurrentUser
│   ├── config/             ← configuration.ts + env.validation.ts
│   ├── http/               ← shared axios client with backend TLS agent; correlation-id middleware; minimal error filter
│   └── health/             ← GET /health (liveness) + GET /health/backend (backend probe)
└── modules/
    ├── auth/               ← forwarded @Public journey routes (login/OTP/MPIN/refresh — throttled) + /healthcheck
    └── proxy/              ← @All('*') wildcard → ProxyService (byte-for-byte relay; 502/504 only on network failure)
```

7. Application Startup Flow

Backend (`HMC_BackEnd`)

```
node dist/main.js
   ↓
main.ts bootstrap()
   ├─ NestFactory.create(AppModule)          ← builds the DI container
   │     ↓
   │  CoreModule loads .env → Joi validation ← BAD ENV = APP REFUSES TO START (fail fast)
   │     ↓
   │  OracleModule / MssqlModule onModuleInit():
   │     ├─ Oracle pool created  (skipped if ORACLE_DISABLED=true)
   │     ├─ Users DB pool created (skipped if USERS_DB_DISABLED=true or host missing)
   │     └─ MOTC SMS DB pool created (skipped if MOTC_SMS_DB_DISABLED=true or host missing)
   │        ⚠ a pool that has credentials but CANNOT connect THROWS → boot fails (by design)
   ├─ body-parser limits set to 15mb (base64 attachments in submits)
   ├─ helmet(), compression(), CORS
   ├─ setGlobalPrefix('api/v1')
   ├─ Swagger document built at /docs (diagnostics paths stripped if DIAGNOSTICS_ENABLED=false)
   └─ app.listen(PORT)
   ↓
log: "Sanaad backend listening on http://localhost:<PORT>/api/v1 (Swagger: /docs)"
```

The boot log tells you exactly which pools were created, skipped, or failed, and prints a connectivity probe for each — **always read the boot log first when something is wrong.**

Gateway (`HMC_Gateway`)

Same shape, no databases: config validation → shared axios client (with optional CA cert for an https backend) → guards/filters → listen on `GATEWAY_PORT`. Route registration order guarantees the literal `/auth/*`, `/health*` routes win over the `@All('*')` wildcard.

---

8. Module-by-Module Documentation

> All endpoints below are live-verified from the generated Swagger document (106 paths). Every route is prefixed with `/api/v1`. **Auth column:** 🔓 = `@Public` (no token), 🔒 = bearer JWT required. The general flow for every module is: `Client → Controller (DTO validation) → Service → Repository port → Oracle/SQL adapter → mapped response → Sanaad envelope`.

8.1 auth — the Sanaad authentication journey (+ sessions)

Location: `src/modules/auth` · **External systems:** Users DB (SQL Server), MOTC SMS DB, LDAP/Entra directory. No Oracle.

Method	Endpoint	Description	Auth
POST	/healthcheck	API-1 — app-launch check (downtime + forced update)	🔓
POST	/auth/initiate	API-2 — validate user, resolve phone, **send OTP**	🔓
POST	/auth/otp/validate	API-3 — validate OTP	🔓
POST	/auth/mpin/update	API-4 — set MPIN (first time)	🔓
POST	/auth/login	API-5 — MPIN login → **JWT + refresh token + functionaccesslist**	🔓
POST	/auth/mpin/forgot	API-6 — forgot MPIN (send OTP)	🔓
POST	/auth/mpin/update/reset	API-7 — reset MPIN with OTP	🔓
POST	/auth/token/refresh	exchange refresh token for a new pair (rotated, one-time use)	🔓
POST	/auth/logout	revoke current access token (+ optional refresh token)	🔒
GET	/auth/me	current identity from the JWT	🔒

Key implementation facts:

- Everything is a **port** with swappable adapters (see §9 and §11): directory (`AUTH_DIRECTORY`), OTP store (`OTP_STORE`), MPIN/device store, function-access source.
- OTP (default `motc` store): generated in code, INSERTed into `MOTC_SMS_PushTable` — **the insert IS the SMS send** (government gateway polls the table). Validation reads the same table back. `MessageID` doubles as the mobile `requestid`.
- MPIN is stored **as received** (the client pre-hashes) and compared by SQL equality — a deliberate legacy-compatibility decision; do not "fix" it to bcrypt/scrypt without a migration plan.
- Testing switches: `AUTH_DISABLED=true` (full dev bypass, any OTP/MPIN accepted) and `AUTH_STATIC_LOGIN=true` (login returns a pinned real AIBRAHIM39 payload and embeds the full user data in the JWT as a `userdata` claim).

8.2 profile — ops 2, 48, 63

Location: `src/modules/profile` · **Oracle:** PERSONAL_DETAILS_V, EMP_PHONE_V, EMP_IN/OUT_ADDRESS_V, EMP_CONTACT_V, DEP_PHONE_V, DEP_ADDRESS_V, UPD_PERSONAL_INFO_PR, EMP_MARITAL_LOV

Method	Endpoint	Description	Auth
GET	/profile?enum=&lang=	op 2 — full personal detail (5 views + dependent child rows in parallel)	🔒
POST	/profile/personal	op 48 — update personal details	🔒
GET	/profile/lov/marital-status	op 63 — marital status LOV	🔒

Special: `addressType` / `addressTypeAr` are intentionally NOT collapsed by localization (client request) — see `PRESERVED_TWIN_BASES` in `shared/utils/localize.util.ts`.

8.3 employee — ops 3, 7, 8, 35, 36

Location: `src/modules/employee` · **Oracle:** EMPLOYMENT_DETAILS_V, SALARY_V, EMPLOYMENT_V, PERFORMANCE_V, SUPERVISOR_VIEW (table function), SUPERVISOR_PR

Method	Endpoint	Description	Auth
GET	/employee/employment	op 3 — employment details + salary/assignment history	🔒
GET	/employee/basic	op 8 — basic info	🔒
GET	/employee/performance	op 7 — performance records	🔒

Method	Endpoint	Description	Auth
GET	/employee/supervisor/views	op 35 — supervisor candidates (returns PERSON_ID!)	🔒
POST	/employee/supervisor	op 36 — change supervisor (`p_new_supervisor` = PERSON_ID, not employee number)	🔒

8.4 payslip — ops 5, 6, 11

Location: `src/modules/payslip` · **Oracle:** GET_PAYSLIP_PERIODS, CHK_PAYROLL_CNT, PAYSLIP_PR

Method	Endpoint	Description	Auth
GET	/payslip/periods	op 5 — pay periods	🔒
GET	/payslip/count	op 6 — payslip count for a period	🔒
GET	/payslip	op 11 — full payslip (PAYSLIP_PR returns **7 REF CURSORs** — earnings, deductions, housing…)	🔒

8.5 leave — ops 9, 10, 12-14, 45-47, 55-58, 61, 62

Location: `src/modules/leave` · **Oracle:** LEAVE_BALANCE_PR, LEAV_OF_ABSEN_NEW_PR, CALC_LEAV_DUR_PR, HR_LEAV_AMEND_PR, HR_LEAV_CANCEL_PR, RET_FRM_LEAV_PR, ABSENCE_V + ~13 LOV views

Method	Endpoint	Description	Auth
GET	/leave/balance?username=&lang=&effectivedate=	op 9 — leave balance (`p_user_name` bound to the request value as-is; legacy `person_id` alias accepted)	🔒
POST	/leave/apply	op 10 — leave submission (LeaveApplyBinds builder, attachments as base64 → BLOB)	🔒
POST	/leave/calculate	op 47 — duration calculation	🔒
POST	/leave/amend · /leave/cancel · /leave/return	ops 57/58/56 — submits	🔒
GET	/leaves?user_name=&leave_type=	leave history (ABSENCE_V)	🔒
GET	/leave/lov/types · reasons · classes · defaults · request-lov · return · return-details · return-related1 · return-related2	ops 12/13/14/45/46/55 + RFL LOVs	🔒
GET	/leave/lov/cancel · /leave/lov/amend (`?person_id=` or `?username=`, optional `?leave_type=` filter on the view's NAME column)	ops 61/62 — the EXACT strings the cancel/amend submits accept	🔒

Gotchas pinned in code comments: none of the leave procedures accept `p_language`; op 61/62 `used_value` is the only text the submit procedures accept; `RET_FRM_LEAV_PR.p_leave_details` is compacted to fit VARCHAR2(60).

8.6 letters — ops 16, 17

Location: `src/modules/letters` · **Oracle:** 7 LOV views + HR_EMPLYMNT_LTR_PR

Method	Endpoint	Description	Auth
GET	/letters/lov	op 16 — all letter-request LOVs (parallel fan-out)	🔒
POST	/letters/apply	op 17 — submit letter request	🔒

8.7 identity — ops 18, 19, 53b, 54, 59, 60

Location: `src/modules/identity` · **Oracle:** QID_DET_V, QID_CHG_PR, COID_REQ_PR, SIT_WORK_LOC_V, SIT_DELEV_LOC_V, SIT_REASON_V

Method	Endpoint	Description	Auth
GET	/identity/qid	op 18 — QID details	🔒
POST	/identity/qid/update	op 19 — QID update	🔒
POST	/identity/idcard/apply	op 54 — company ID request	🔒
GET	/identity/lov/work-location · delivery-location · reason	ops 53b/59/60	🔒

8.8 contact — ops 25, 27-30, 32

Location: `src/modules/contact` · **Oracle:** PHONE_TYPE_V, COUNTRY_LOV, PHONE_PKG.ADD_OR_UPDATE_PHONE, DEL_PHONE_NUMBER_PR, CREATE_ADDRESS_PR, UPD_ADDRESS_PR

Method	Endpoint	Description	Auth
GET	/contact/lov/phone-type · /contact/lov/country	ops 27/30	🔒
POST	/contact/phone	op 28 — upsert phones (submitted per-phone; stops at first failure)	🔒
POST	/contact/phone/delete	op 32	🔒
POST	/contact/address · /contact/address/update	ops 29/25 — `p_country` = country NAME ("Qatar"); date-tracked (same-day repeat update fails)	🔒

8.9 dependents — ops 24, 31, 33, 34, 49, 64, 65

Location: `src/modules/dependents` · **Oracle:** ADD_DEPENDENT_PKG (add/update), REMOVE_DEPENDENT_PR, PASS_DTL_PR, DEP_LOOKUP_LOV, DEP_PLACE_LOV, PASSPORT_TYPE

Method	Endpoint	Description	Auth
POST	/dependents	op 65 — add dependent (strict flexfield rules: ≥1 attachment, passport fields, unique QID, relationship from op-64 CONTACT group)	🔒
POST	/dependents/update · /dependents/delete	ops 24/31	🔒
GET	/dependents/lov	op 64 — multi-type LOV (also serves ADDRESS_TYPE/SPONSORSHIP/VISA groups)	🔒
GET	/dependents/passport/types · /dependents/passport/issue-place	ops 33/49	🔒
POST	/dependents/passport/apply	op 34	🔒

Accepts the legacy misspelled `p_*` names (`p_gendar` , `p_relation_ship` , `p_visa_validy` , …) and mirrors them to canonical ones.

8.10 school-fees — ops 37-40, 50, 52, 53

Location: `src/modules/school-fees` · **Oracle:** SCHOOL_FEE_PR, CHILD_DETS_VIEW (table function), SCHOOL_NAME_LOV (+4 more LOVs)

Method	Endpoint	Description	Auth
POST	/school-fees/apply	op 39 — school-fee request	🔒
GET	/school-fees/lov/schools (search/page/pageSize) · terms · edu-stage · academic-year · request-type	ops 37/38/40/50/53	🔒
GET	/school-fees/children	op 52 — child details	🔒

8.11 appointments — ops 41-44 (Cerner, NOT Oracle)

Location: `src/modules/appointments` · **External:** Cerner HTTP (`CERNER_BASE_URL`)

Method	Endpoint	Description	Auth
GET	/appointments/upcoming · /appointments/masters · /appointments/booking-init	ops 41/42/43	🔒
POST	/appointments/book	op 44	🔒

⚠ Returns **503** when `CERNER_BASE_URL` is unset (current staging state). This is environment configuration, not a code bug — do not work around it in code.

8.12 annual-ticket — ops 66, 67, 72

Location: `src/modules/annual-ticket` · **Oracle:** ANNUAL_TICKT_LOV, TICKET_REQ_PR, CANCEL_TKT_PR + 3 cancellation views

Method	Endpoint	Description	Auth
GET	/annual-ticket/master	op 66	🔒
POST	/annual-ticket/apply	op 67 — `p_employee` MUST be the Oracle PERSON_ID; `p_contractual_year` must exist in HMC_HR_CONTRACTUAL_YEAR_SIT	🔒
GET	/annual-ticket/cancel-options	op 72 — tickets + takenAs + repayment methods (parallel)	🔒
POST	/annual-ticket/cancel	op 72	🔒

8.13 approvals — ops 20-23, 68-71 + RFMI

Location: `src/modules/approvals` · **Oracle:** APPROVE_SUMRY_V, MY_REQEST_SUMMARY_V, PNDNG_QID_V, WORKLISTS_V, ACTION_HISTORY_V, NOTYFY_APPR_V, HR_ATTACHMENTS_V, ~10 per-service PNDNG_* detail views (allow-listed), APPROVE_REJECT_PR, HR_RFMI_PR, REASSIGN_PR

Method	Endpoint	Description	Auth
GET	/approvals	op 20 — approvals summary	🔐
GET	/approvals/my-requests	op 23	🔐
GET	/approvals/worklist · /approvals/worklist/summary · /approvals/worklist/{id}/history	ops 68/69/70 (keyed by USERNAME, not employee number)	🔐
GET	/approvals/{id}/details · /approvals/attachments/{documentId}	op 21 + attachment download	🔐
POST	/approvals/{id}/decision · /approvals/{id}/request-info · /approvals/{id}/reassign	ops 22 / RFMI / 71	🔐

8.14 lookups (shared kernel) — ops 15, 26 + generic reads

Location: `src/lookups`

Method	Endpoint	Description	Auth
GET	/lookups/yes-no · /lookups/rfmi-user	ops 15/26	🔐
GET	/lookups/lov?lovname= · /lookups/master?lookupname=	generic reads via the registry in `shared/constants/lov-names.ts`	🔐

All feature-module LOV endpoints internally call `LookupsService.getByObject(...)` — LOV reads are cached per `(object, lang, username, options)` for `LOV_CACHE_TTL_MS` (default 5 min) with concurrent-read coalescing.

8.15 Operational endpoints (not for the mobile app)

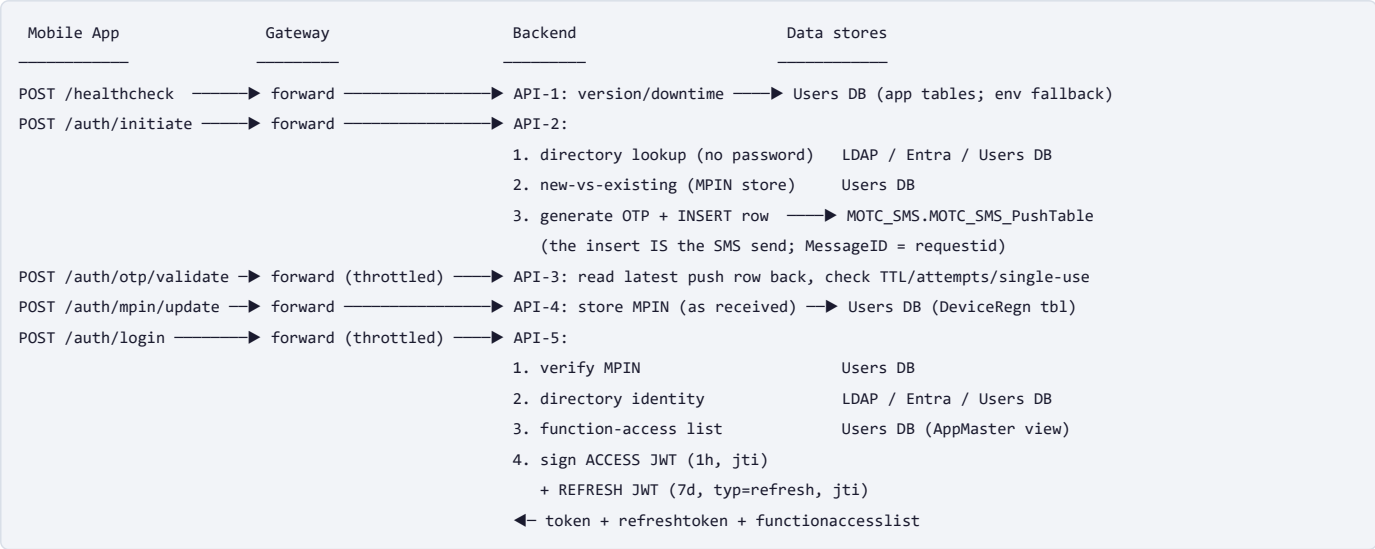
Group	Endpoints	Gate
Health	GET /health 🔐 · /health/db · /health/users-db · /health/motc-sms-db	DB probes gated by `DIAGNOSTICS_ENABLED`
Oracle diagnostics	GET /diagnostics/oracle-object?name= · /diagnostics/oracle-logs (+/stats, DELETE, /view HTML)	`DIAGNOSTICS_ENABLED`
SQL consoles	POST /diagnostics/users-db/sql · /diagnostics/motc-sms-db/sql (SELECT-only)	`USERS_DB_SQL_ENABLED` / `MOTC_SMS_SQL_ENABLED`, always 403 in production
API logs	GET /api-logs (+/errors, /success, /slow, /statistics, /{id}, /view HTML, DELETE)	`DIAGNOSTICS_ENABLED`
Dev console	GET /dev-console (SQL worksheet + API tester)	`DEV_CONSOLE_ENABLED`, optional token, disabled in production

8.16 Gateway routes

Method	Endpoint	Behavior	Auth
POST	/healthcheck, /auth/initiate, /auth/mpin/update	forwarded verbatim	🔓
POST	/auth/otp/validate, /auth/login, /auth/mpin/forgot, /auth/mpin/update/reset, /auth/token/refresh	forwarded verbatim + **throttled** (default 5/min per IP)	🔓
GET	/health · /health/backend	gateway liveness · backend probe	🔓
ANY	everything else (`@All('*')`)	JWT validated locally → proxied byte-for-byte	🔐

9. Authentication & Authorization

9.1 The login journey (what the mobile app does)



9.2 After login

- The app sends `Authorization: Bearer <access token>` on every call.
- The **gateway** verifies the signature **locally** (shared `JWT_SECRET` — same value in both `.env`s, must stay in sync) and proxies the request. It never calls the backend to validate.
- The **backend** verifies again (`JwtStrategy`), additionally rejecting: refresh tokens used as access tokens (`typ=refresh`), and tokens whose `jti` was revoked by logout (in-memory `TokenRevocationService`). **The backend is the enforcement point for logout.**
- `POST /auth/token/refresh` (public, throttled) exchanges a valid refresh token for a fresh pair; the used refresh token is revoked (one-time use / rotation).
- `POST /auth/logout` revokes the current access token's `jti` (+ the refresh token if sent in the body).

JWT claims: `sub` (employee number or username), `username`, `employeeNumber`, `roles` (e.g. `["EMPLOYEE"]`), `functions` (enabled function codes), `name`, `dept`, `company`, `jti`, `iat`, `exp` — plus a full `userdata` object when `AUTH_STATIC_LOGIN=true`.

9.3 Authorization

- **RolesGuard** — routes decorated with `@Roles(Role.APPROVER, ...)` require that role in the token (approvals module).
- **FunctionAccessGuard** — routes decorated with `@RequireFunction('frmSchoolFees')` require that function code in the token's `functions` claim (which came from the Users-DB AppMaster view at login).
- Undecorated routes need only a valid token.

9.4 Switchable providers (all decided by env, no redeploy)

Concern	Env switch	Options
Employee directory	`AUTH_DIRECTORY`	`ldap` (LDAPS/ldapts) · `entra` (Microsoft Graph app-only) · `usersdb` (legacy device-table only, no corporate directory)
OTP store + delivery	`OTP_STORE`	`motc` (default, MOTC_SMS_PushTable) · `legacy` (HMC_RHAP_OTP_tbl + HTTP SMS adapter)
Everything (dev)	`AUTH_DISABLED=true`	full bypass: guards permissive, synthesized identities, any well-formed OTP/MPIN accepted
Static login (testing)	`AUTH_STATIC_LOGIN=true`	login returns a pinned real payload; full user data embedded in the JWT; rest of API enforces tokens normally

9.5 Security invariants (do not break these)

- Raw OTPs and unmasked phone numbers are **never logged** (both SQL services redact; the SMS adapter masks phones).
- MPIN values are redacted from logs and stored as received (legacy decision — see §21 before changing).

- SQL is always parameterized (`:binds` in Oracle, `@params` in SQL Server). Object/table names interpolated into SQL are validated against allow-lists (`ORACLE_OBJECTS` , identifier regexes).
- Known limitation: token revocation is in-memory — a backend restart forgets logouts (until token expiry), and multiple instances don't share the denylist. Accepted trade-off; documented in `TokenRevocationService` .

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10. Database Architecture

There is no ORM, no entities, no migrations — on purpose. All schemas are owned by other teams (Oracle HR, legacy Sanaad SQL Server, MOTC). This project only *reads views* and *calls procedures*, with parameterized SQL.

10.1 The three databases

Database	Driver/Pool	What lives there	Service
Oracle HR (`XXHMC_SND_*` schema)	node-oracledb pool	~95 allow-listed views/LOVs/procedures implementing all 71 business ops	`core/database/oracle.service.ts`
Users/Sanaad DB (SQL Server)	mssql pool #1 (`USERS_DB_*`)	`HMC_Sanad_DeviceRegn_tbl` (device+MPIN), `HMC_RHAP_OTP_tbl` (legacy OTP), app master/downtime/update tables, `HMC_Sanad_AppMaster_VW` (function access)	`core/database/mssql.service.ts`
MOTC_SMS DB (SQL Server, named instance, static port 9001)	mssql pool #2 (`MOTC_SMS_DB_*`)	`MOTC_SMS_PushTable` — the government SMS outbox; OTP rows are stored AND delivered through it	`core/database/motc-sms-db.service.ts`

10.2 How Oracle access works (the most important pattern in the codebase)

Problem this project solved: the legacy documentation describes *request parameters*, not the actual database objects — column names and procedure signatures differ per object. Guessing produced `ORA-00904`, `PLS-00306`, etc.

Solution — **ask the database itself** (`OracleSchemaService` reads the data dictionary) and centralize the calling patterns in `BaseOracleRepository`, which every Oracle adapter extends:

Helper	Use case	Behavior
`query(sql, binds)`	plain view SELECT	parameterized
`readByResolvedKey(view, value, candidates)`	"filter this view by user" when the key column name varies (`USER_NAME` vs `USERNAME` vs `EMPLOYEE_NUMBER` ...)	resolves the real column from ALL_TAB_COLUMNS; on schema mismatch logs `SCHEMA_MISMATCH` and returns `[]` instead of failing the request
`readByResolvedKeyAny / In`	same, matching several identifier forms / IN-lists	chunked, parameterized
`callSubmitProc(proc, params, values)`	business submits (`*_PR`)	resolves the DECLARED argument list (incl. overloads) from ALL_ARGUMENTS, binds by name, maps OUT params (`p_success_flag/p_error_msg[_ar]`) to a `SubmitResult`
`callRowsProc(proc, params, values)`	procedures returning a REF CURSOR	finds the cursor parameter's real name
`callMultiCursorProc`	PAYSLIP_PR's 7 cursors	reads all cursors on one connection
`callRowsOrTableFunction`	object may be a proc OR a table function	dictionary decides; `SELECT * FROM TABLE(fn(...))` fallback

Every single Oracle call is recorded (SQL, sanitized binds, duration, rows, ORA code) in an in-memory store — inspect via `GET /diagnostics/oracle-logs` or the HTML view. This is your #1 debugging tool.

Allow-list rule: any Oracle object name must exist in `shared/constants/oracle-objects.ts` before it can be queried — this is both an injection defense and a catalog.

10.3 SQL Server access

Both `MssqlService` and `MotcSmsDbService` expose only parameterized `query` / `execute` (named `@params`), plus `ping()` / `diagnose()`. Sensitive params (mpin/otp/password/messagebody/phone) are redacted from logs. Pools are created at boot; `*_DISABLED=true` skips creation (routes then fail with a typed 503 while the rest of the API works).

10.4 Data flow summary

```
Read op:    Controller → Service → OracleRepository.readByResolvedKey(VIEW) → rows
            → mapper (col/str/dateStr, URL-decode Arabic) → ResponseInterceptor
            → localizeArTwins (EN/AR twin collapsing per ?lang=) → Sanaad envelope → client

Submit op:  Controller (strict DTO) → Service → OracleRepository.callSubmitProc(PROC)
            → OUT params → SubmitResult {successflag, message, messageAr}
            → HTTP 200 ALWAYS (Sanaad convention: business result is successflag, not the status code)
```

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11. External Integrations

11.1 Oracle HR (business core) — see §10.2.

11.2 Users/Sanaad SQL Server — auth cycle + API-1

Config: `USERS_DB_*`. Implemented in `modules/auth/infrastructure/adapters/mssql-*.ts`. Failure mode: typed 503 per call; healthcheck falls back to `APP_*` env config.

11.3 MOTC SMS gateway DB — OTP store + SMS delivery

Config: `MOTC_SMS_DB_*` + fixed INSERT column values (`MOTC_SMS_APP_ID`, `MOTC_SMS_SUBJECT_ID`, priority/language/etc. — several still pending from the client). Implemented in `modules/auth/infrastructure/adapters/motc-sms-otp.repository.ts` (+ `core/database/motc-sms-db.service.ts`). `MessageID` is generated as MAX+1 with duplicate-key retry. `BusinessParam1/2` carry username+IMEI by default (what makes per-user OTP validation possible). The OTP text template is `SMS_MESSAGE_TEMPLATE` (`{otp}` placeholder).

11.4 LDAP / Active Directory (LDAPS)

Config: `LDAP_*` (host, base DN, bind account, CA cert inline or path). Implemented in `ldap-user.repository.ts` (ldapts). Only `validate()` (passwordless identity+phone lookup) is used by the journey. TLS: provide `LDAP_CA_CERT[_PATH]` and set `LDAP_TLS_REJECT_UNAUTHORIZED=true` in production.

11.5 Microsoft Entra ID (Graph) — alternative directory

Config: `ENTRA_*` (tenant, client id/secret — app needs `User.Read.All` application permission with admin consent). Implemented in `entra-graph-user.repository.ts` (client-credentials token + Graph user lookup). Selected with `AUTH_DIRECTORY=entra`.

11.6 Cerner — staff-clinic appointments

Config: `CERNER_BASE_URL`, `CERNER_TIMEOUT_MS`. Implemented in `modules/appointments/infrastructure/cerner/` (`CernerClient` refuses to call when unconfigured → clean 503). **Requires Knowledge Transfer:** the real Cerner endpoint/contract for each environment.

11.7 Legacy HTTP SMS gateway (only when `OTP_STORE=legacy`)

Config: `SMS_API_*`. Implemented in `sms-otp-delivery.adapter.ts` — a generic JSON POST placeholder awaiting the corporate contract. Unset base URL = masked log-only in dev, hard 503 in production.

Error-handling rule for ALL integrations: adapter throws a typed exception → `AllExceptionsFilter` classifies it → client gets a safe, generic message; the full detail goes to server logs / api-logs / oracle-logs.

---

12. Configuration & Environment Variables

Both projects: `.env` (never committed) → Joi schema validates at boot (bad env = refusal to start) → typed namespaces in `core/config/configuration.ts` → injected via `ConfigService`.

12.1 Backend (HMC_BackEnd) — grouped; placeholders only

Variable	Required	Purpose	Example
NODE_ENV	no (default development)	environment mode; production hardens several features	`production`
PORT / API_PREFIX / CORS_ORIGINS	no	listener + prefix + CORS	`3009` / `api/v1` / `*`
REQUEST_TIMEOUT_MS	no	global request timeout (408)	`30000`
LOG_LEVEL	no	error/warn/log/debug/verbose	`debug`
Oracle ORACLE_USER / ORACLE_PASSWORD / ORACLE_DSN	yes (unless disabled)	pool credentials + EZConnect DSN	`xxhmc_snd` / `YOUR_PASSWORD` / `host:1521/SERVICE`
ORACLE_POOL_MIN/MAX, ORACLE_QUEUE_TIMEOUT_MS, ORACLE_CALL_TIMEOUT_MS	no	pool + per-call timeouts	`2` / `10` / `25000` / `25000`
ORACLE_DISABLED / ORACLE_THICK_MODE / ORACLE_CLIENT_LIB_DIR	no	skip pool · thick mode + Instant Client path	`false` / `true` /``
Users DB USERS_DB_HOST/PORT/NAME/USER/PASSWORD	yes (unless disabled)	SQL Server #1	`sqlhost.hmc.org.qa` / `1433` / `...` / `YOUR_PASSWORD`
USERS_DB_ENCRYPT / USERS_DB_TRUST_SERVER_CERT / USERS_DB_DISABLED / USERS_DB_SQL_ENABLED	no	TLS · self-signed trust · skip pool · SQL console	`true` / `false` / `false` / `false`
MOTC SMS DB MOTC_SMS_DB_HOST/PORT/NAME/USER/PASSWORD	yes for OTP (unless disabled)	SQL Server #2 (named instance + static port)	`HSHCL7VVSQ1\SQL1` / `9001` / `MOTC_SMS` / `Sanaad_Liv_Usr` / `YOUR_PASSWORD`
MOTC_SMS_DB_ENCRYPT / MOTC_SMS_DB_TRUST_SERVER_CERT / MOTC_SMS_DB_DISABLED / MOTC_SMS_SQL_ENABLED	no	TLS · trust self-signed · skip pool · SQL console	`true` / `true` / `false` / `false`
MOTC_SMS_TABLE, MOTC_SMS_APP_ID, MOTC_SMS_FROM_ADDRESS, MOTC_SMS_SUBJECT_ID, MOTC_SMS_PRIORITY, MOTC_SMS_LANGUAGE_ID, MOTC_SMS_RECIPIENT_ADDRESS_TYPE, MOTC_SMS_PROCESSED_STATE, MOTC_SMS_MESSAGE_EXPIRE_MINUTES, MOTC_SMS_CUSTOMER_ID, MOTC_SMS_MASK_MESSAGE_LOG, MOTC_SMS_BUSINESS_PARAM1/2	APP_ID+SUBJECT pending from client	fixed columns of the documented INSERT	see `.env.example`
OTP_STORE	no	`motc` (default) or `legacy`	`motc`
OTP_LENGTH / OTP_TTL_SECONDS / OTP_MAX_ATTEMPTS / OTP_RESEND_WINDOW_SECONDS	no	OTP policy	`6` / `300` / `5` / `60`
SMS (legacy) SMS_API_BASE_URL/KEY, SMS_SENDER_ID, SMS_MESSAGE_TEMPLATE	only for legacy store	HTTP SMS adapter · OTP text (`{otp}`)	`https://...` / `YOUR_KEY` / `HMC` / `Your Sanaad verification code is {otp}`
JWT JWT_SECRET / JWT_ISSUER / JWT_AUDIENCE / JWT_EXPIRES_IN / JWT_REFRESH_EXPIRES_IN	SECRET: yes	token signing — **must match the gateway**	`YOUR_LONG_RANDOM_SECRET` / `sanaad` / `sanaad-b2e` / `1h` / `7d`
AUTH_DISABLED / AUTH_STATIC_LOGIN	no (never true in prod)	dev bypass · static test login	`false` / `false`
AUTH_DIRECTORY	no	`ldap` · `entra` · `usersdb`	`usersdb`
FUNCTION_ACCESS_VIEW / FUNCTION_ACCESS_APP_ID	no	login function list source	`HMC_Sanad_AppMaster_VW` / `1`
MPIN MPIN_MIN/MAX_LENGTH, MPIN_MAX_ATTEMPTS, MPIN_LOCKOUT_MINUTES	no	MPIN policy	`4` / `6` / `5` / `15`

Variable	Required	Purpose	Example
LDAP LDAP_ENABLED, LDAP_HOST/PORT, LDAP_BASE_DN, LDAP_SEARCH_FILTER, LDAP_BIND_DN/PASSWORD, LDAP_CA_CERT[_PATH], LDAP_TLS_REJECT_UNAUTHORIZED, LDAP_TIMEOUT_MS	when AUTH_DIRECTORY=ldap	directory lookup	see `.env.example`
Entra ENTRA_TENANT_ID / ENTRA_CLIENT_ID / ENTRA_CLIENT_SECRET (+ URLs, lookup attr)	when AUTH_DIRECTORY=entra	Graph lookup	`YOUR_TENANT` / `YOUR_CLIENT_ID` / `YOUR_SECRET`
Cerner CERNER_BASE_URL / CERNER_TIMEOUT_MS	for appointments	Cerner service	`https://cerner.example.hamad.qa` / `10000`
App launch APP_NAME, APP_MIN_SUPPORTED_VERSION, APP_LATEST_VERSION, APP_DOWNTIME(_START/_END)	no	API-1 fallback config	`SanaadHealth` …
DIAGNOSTICS_ENABLED	no	master switch for /diagnostics, /api-logs, DB health tests + Swagger stripping	`true`
DEV_CONSOLE_ENABLED / DEV_CONSOLE_TOKEN / DEV_CONSOLE_ALLOW_WRITE	no	dev console gate	`true` /`` / `false`
LOV_CACHE_TTL_MS	no	LOV cache lifetime	`300000`

12.2 Gateway (HMC_Gateway)

Variable	Required	Purpose	Example
GATEWAY_PORT / API_PREFIX / CORS_ORIGINS	no	listener	`3001` / `api/v1` / `*`
BACKEND_BASE_URL / BACKEND_API_PREFIX	yes	upstream backend	`http://localhost:3009` / `api/v1`
BACKEND_TIMEOUT_MS	no	proxy timeout — keep it ABOVE the backend's REQUEST_TIMEOUT_MS (e.g. 35000 vs 30000) so backend errors win over generic 504s	`35000`
BACKEND_CA_CERT / BACKEND_CA_CERT_PATH / BACKEND_TLS_REJECT_UNAUTHORIZED	for https backend	TLS trust	`/` / `true`
JWT_SECRET / JWT_ISSUER / JWT_AUDIENCE	yes	**must equal the backend's values**	`YOUR_LONG_RANDOM_SECRET`
AUTH_DISABLED	no (never true in prod)	bypass gateway JWT check	`false`
THROTTLE_LOGIN_LIMIT / THROTTLE_LOGIN_TTL_MS	no	rate limit for login surfaces	`5` / `60000`
REQUEST_TIMEOUT_MS / LOG_LEVEL / NODE_ENV	no	misc	`35000` / `debug` / `development`

13. API Request Lifecycle

An authenticated business request (`GET /api/v1/leave/balance`) end to end:

```
Mobile client
 ↓  Authorization: Bearer <JWT>
GATEWAY
 ↓ CorrelationIdMiddleware        (x-correlation-id created/propagated)
 ↓ JwtAuthGuard                   (local signature+expiry check — 401 here if bad)
 ↓ ProxyController @All('*') → ProxyService
 ↓   forwards method/path/query/body + authorization/content-type/accept/accept-language
BACKEND
 ↓ body parsed (15mb limit) → CorrelationIdMiddleware
 ↓ ApiLogInterceptor  (outermost — records everything, success or failure)
 ↓ AuditInterceptor   (level-1 API_CALL audit)
 ↓ TimeoutInterceptor (starts the 30s clock → 408)
 ↓ ResponseInterceptor (waits to wrap the result on the way OUT)
 ↓ JwtAuthGuard        (re-verifies; rejects revoked jti / refresh-typ tokens)
 ↓ RolesGuard → FunctionAccessGuard (only if the route is decorated)
 ↓ ValidationPipe      (DTO validation; unknown keys → 400)
 ↓ LeaveController.balance(q)
 ↓ LeaveService.getBalance(query)
 ↓ LeaveOracleRepository → BaseOracleRepository.callRowsProc(LEAVE_BALANCE_PR)
 ↓ OracleService.callCursor  → Oracle executes, REF CURSOR fetched
 ↓   (the call is recorded in OracleLogStore with sql/binds/duration)
 ↑ rows → LovMapper/mappers → service → controller returns plain data
 ↑ ResponseInterceptor: localizeArTwins(lang) + wrap → { result, opstatus, status, httpStatusCode }
 ↑ ApiLogInterceptor completes its record (status, duration)
GATEWAY relays the response byte-for-byte (only network failures become gateway 502/504)
Mobile client
```

Error path: ANY exception thrown anywhere is caught by `AllExceptionsFilter`, classified, logged in full (with correlation id), and returned as a safe envelope: `{ success:false, message, status:'error', httpStatusCode }`.

---

14. Error Handling & Logging

14.1 The error pipeline

1. **Validation errors** — global `ValidationPipe` rejects malformed/unknown fields → 400 with detail list.
2. **Typed domain errors** — adapters throw meaningful exceptions (`OracleQueryError`, `MssqlQueryError`, `MssqlUnavailableException`, `NotFoundException`, …).
3. `AllExceptionsFilter` (backend) catches everything, uses the **exception classifier** (`core/http/exception-classifier.ts`) to bucket it (VALIDATION / DATABASE_ERROR / EXTERNAL_SERVICE / AUTH / TIMEOUT / UNKNOWN), picks a safe bilingual message (`shared/constants/error-codes.ts` maps ORA codes → HTTP statuses), logs the full stack server-side, and returns the consistent error envelope.
4. **Sanaad submit convention** — business submits return **HTTP 200 even on business failure**; the outcome is in `successflag` (`S` / `E`) + `message` / `messageAr`. Don't "fix" this — the mobile app depends on it.
5. Gateway: relays backend errors untouched; only a real network failure produces a gateway-originated `502` (unreachable) or `504` (timeout).

14.2 Where to look when something fails (in order)

Tool	URL / place	What it tells you
Boot log	console	which DB pools were created/skipped/FAILED, with the exact reason
API logs	`GET /api-logs/view` (HTML) or `/api-logs?…`	every request: status, duration, user, error category, stack trace (`/api-logs/{id}`)
Oracle call log	`GET /diagnostics/oracle-logs/view` or JSON	every Oracle call: final SQL, sanitized binds, duration, ORA code, correlation id
Object describe	`GET /diagnostics/oracle-object?name=XXHMC_SND_…`	the REAL columns/parameters of a view/procedure
SQL consoles	`POST /diagnostics/users-db/sql`, `/motc-sms-db/sql`	ad-hoc SELECTs against the SQL Server DBs (dev/staging only)
Dev console	`GET /dev-console`	Oracle SQL worksheet, PL/SQL source browser, API tester that links a request to its Oracle calls
Health probes	`/health`, `/health/db`, `/health/users-db`, `/health/motc-sms-db`	connectivity, latency, server version, exact failure message
Audit log	stdout JSON lines (`AuditService`)	login lifecycle, security incidents

Correlate everything with the `x-correlation-id` response header.

14.3 Debugging examples

- *A submit returns successflag E with a generic message* → open `/diagnostics/oracle-logs`, find the call, read the real `p_error_msg` / ORA code; check the procedure's declared params with `/diagnostics/oracle-object?name=...`.
- *A view read returns `[]` unexpectedly* → look for a `SCHEMA_MISMATCH` warning in the logs: the view doesn't have the key column the code expected (by design it degrades instead of erroring).
- *Login flow fails* → check boot log for Users-DB/MOTC pool state, then `/health/users-db` + `/health/motc-sms-db` for the exact connection error (TLS? login? DNS? see §18).

15. How to Develop a New Feature

Follow the existing architecture — copy the closest existing module rather than inventing a new style. `identity/` or `letters/` are good small templates.

15.1 Add a new module

```
src/modules/<name>/
├── <name>.module.ts
├── interface/<name>.controller.ts + interface/dto/<name>.dto.ts
├── application/<name>.service.ts
├── domain/<name>.repository.ts          ← interface + `export const X_REPOSITORY = Symbol(...)`
└── infrastructure/oracle/<name>.oracle.repository.ts   ← extends BaseOracleRepository
```

1. Declare the module: controller in `controllers`, service + `{ provide: X_REPOSITORY, useClass: XOracleRepository }` in `providers`.
2. Import it in `src/app.module.ts`.
3. If it reads any new Oracle object: **add the name to `shared/constants/oracle-objects.ts` first** (queries against unknown objects are rejected).

15.2 Add a new endpoint to an existing module

1. Add a DTO in `interface/dto/` (extend `LangQueryDto` / `ProfileQueryDto` for reads; use the helpers in `shared/dto/oracle-submit.dto.ts` for `p_*` submit bodies).
2. Add the controller method with `@Get/@Post`, `@ApiOperation({ summary: 'op NN — ...' })`, and `@ApiReadOkResponse` / `@ApiActionOkResponse`.
3. Add the service method; call the repository port — never SQL in the service.
4. Implement the adapter method with the right base helper (`readByResolvedKey` for view reads, `callSubmitProc` for submits, `callRowsProc` for REF-CURSOR procs).
5. Add a `*.spec.ts` beside the adapter mocking `OracleService` / `OracleSchemaService` (copy an existing spec's `make()` pattern).
6. `npm.cmd run build` + `npm.cmd test`, then verify by hand in Swagger `/docs` (or the dev console API tester).

15.3 Add validation

Use class-validator decorators on the DTO — the global pipe does the rest. Remember `forbidNonWhitelisted: true`: every accepted field must be declared. Legacy alternate spellings are handled by declaring both and mirroring in the adapter (see `dependents.binds.ts`).

15.4 Database changes

There are no migrations here — schema changes happen on the Oracle/SQL Server side by their owners. When a procedure/view changes:

1. Inspect the new shape: `GET /diagnostics/oracle-object?name=…`.
2. Usually nothing else is needed — signatures are resolved at runtime. Update the documented param list / DTO only if the *request contract* changed.
3. Record any live-verified behavior in `AGENTS.md` (that file is the team's memory).

15.5 Protect an endpoint

- Default: authenticated by the global guard (do nothing).
- Public: `@Public()` (only for the pre-login journey!).
- Role-restricted: `@Roles(Role.APPROVER)`.
- Function-restricted: `@RequireFunction('frmSchoolFees')`.
- If you add a truly new public route, remember the gateway: it must either be declared there (like the auth journey) or it will require a JWT at the gateway.

---

16. Testing & Debugging

16.1 What exists

- **Backend:** 22 unit suites / ~170 tests, co-located as `src/**/*.spec.ts` (Oracle adapters with mocked pools, OTP stores, auth service, guards, utils). **No e2e specs** (`npm.cmd run test:e2e` runs nothing — the config exists, the specs were never written).
- **Gateway:** 4 unit suites + a real e2e suite (`test/gateway.e2e-spec.ts`) that boots the app against `test/mock-backend.ts` (mints real JWTs) — covers proxying, auth, throttling, 502 mapping.

16.2 Running tests

```
npm.cmd test              # all unit tests
npm.cmd test -- --watch   # watch mode
npm.cmd run test:cov      # coverage report → coverage/
npx.cmd jest src/modules/leave   # one folder/file
npm.cmd run test:e2e      # gateway only (backend has no e2e specs)
```

Tests never need a database — everything external is mocked.

16.3 Debugging in VS Code

1. Run `npm.cmd run start:debug` (starts ts-node-dev with `--inspect`).
2. VS Code → Run and Debug → "Attach to Node Process" (or add a launch config with `"request": "attach", "port": 9229`).
3. Set breakpoints directly in the `.ts` files (source maps work), e.g. in a controller method.
4. Trigger the request from Swagger `/docs` ("Authorize" with a token from `/auth/login`), Postman (`postman/HMC-Sanaad-Full.postman_collection.json` has verified examples), or the dev console API tester.

Debugging a single Jest test: `npx.cmd jest path/to.spec.ts --runInBand` with a `debugger;` statement, launched from VS Code's JavaScript Debug Terminal.

16.4 Debugging an API request without a debugger

Use the built-in observability instead of console.log: `/api-logs/view` for the request, `/diagnostics/oracle-logs/view` for its DB calls (same correlation id), `/dev-console` to replay it.

---

17. Important npm Commands

Same scripts in both projects (run with `npm.cmd run <name>`):

Command	Purpose
`build`	Compile TS → `dist/` (`nest build` + `tsc-alias` for path aliases). Also the typecheck.
`start` / `start:prod`	Run the compiled app (`node dist/main`)
`start:dev`	Development server with hot reload (ts-node-dev)
`start:debug`	Dev server with the Node inspector for VS Code attach
`test` / `test:watch` / `test:cov`	Jest unit tests / watch / coverage
`test:e2e`	e2e tests (gateway has them; backend currently has none)
`lint`	ESLint with `--fix` ⚠ on the backend this crashes (known env issue) — use `npx.cmd eslint src --ext .ts` instead, and ignore the CRLF `Delete ␍` noise
`format`	Prettier ⚠ do NOT run on the backend (CRLF working copy — it would rewrite every file). Gateway is fine.
`depcruise` (backend only)	dependency-rule check (`.dependency-cruiser.js`)

18. Troubleshooting

Symptom	Likely cause → fix
`npm : File ...npm.ps1 cannot be loaded`	PowerShell execution policy → always use **`npm.cmd` / `npx.cmd`**
App refuses to start with a Joi message	invalid/missing env var — the message names it; fix `.env`
Boot fails: `Failed to create <X> DB pool`	that DB is unreachable/misconfigured. Boot **intentionally** fails. Temporarily set `ORACLE_DISABLED` / `USERS_DB_DISABLED` / `MOTC_SMS_DB_DISABLED` =true to boot without it
`...self-signed certificate` (SQL Server)	set `USERS_DB_TRUST_SERVER_CERT=true` / `MOTC_SMS_DB_TRUST_SERVER_CERT=true` (internal instances use self-signed certs)
`Login failed for user '...'` (SQL Server)	credential/permission issue — verify password (quote values containing `#` in `.env`), then ask the DB team for the 18456 state code (5=no login, 8=bad password, 38=no DB access, 58=Windows-auth-only)
`getaddrinfo ENOTFOUND <shortname>` in Docker	Linux containers can't resolve short Windows hostnames → use FQDN/IP or `extra_hosts`
`EADDRINUSE` port already in use	another instance running → change `PORT` / `GATEWAY_PORT` or kill the process (`netstat -ano
Gateway returns `{"httpStatusCode":504}`	backend didn't answer within `BACKEND_TIMEOUT_MS` — check `GET /health/backend`, backend boot state, and which dependency hangs (`/api-logs/slow`)
401 on everything	missing/expired/revoked token, or `JWT_SECRET` mismatch between gateway and backend (they MUST be identical)
OTP endpoints return 503	MOTC pool not created (boot log) or `OTP_STORE=legacy` without SMS config
Appointments return 503	`CERNER_BASE_URL` not configured — environment issue, do not patch code
Oracle `MODULE_NOT_FOUND` from `dist/`	stale incremental build artifact → delete `tsconfig.build.tsbuildinfo`, rebuild (and never re-add `"incremental": true`)
ESLint `TypeError: expand is not a function`	known env issue → `npx.cmd eslint src --ext .ts`
Thousands of `Delete ␍` lint errors	CRLF working copy — filter them out; do not run prettier `--fix` on the backend
400 `property X should not exist`	`forbidNonWhitelisted` — add the field to the DTO (or fix the client payload)
413 / body too large	body limit is 15mb (`BODY_LIMIT` in `main.ts`) — attachments are base64 (~4/3 inflation)

19. Learning Roadmap (First Week)

Day 1 — run it. Install Node 20 + VS Code. Clone. `npm.cmd ci` in both projects. Do the §3.5 no-database smoke test. Open `/docs`, `/health`, `/api-logs/view`. Read `AGENTS.md` top to bottom (skim is fine).

Day 2 — language basics + bootstrap. Work through §4 with the files open: `package.json`, `HMC_BackEnd/src/main.ts`, `core/config/configuration.ts` + `env.validation.ts`. Change a default (e.g. PORT), watch Joi reject a bad value.

Day 3 — one full request. With `start:dev` running and §13 next to you, follow `GET /api/v1/lookups/yes-no?lang=en` from Swagger through `lookups.controller.ts → lookups.service.ts → lov.oracle.repository.ts` with breakpoints (or the log pages). Then read `core/core.module.ts` (pipeline order) and `core/http/response.interceptor.ts`.

Day 4 — auth + data layer. Read §9, then `modules/auth/application/auth.service.ts`, `core/auth/jwt.strategy.ts`, and the port/adapter pairs in `modules/auth`. Then §10 + `core/database/base.repository.ts` (the helpers) and one adapter (`leave.oracle.repository.ts`). Log in via Swagger (`AUTH_DISABLED=true`), decode your JWT at jwt.io.

Day 5 — make a change. Pick something small: add a query filter to an existing LOV endpoint (mirror §15.2), write its spec, run build+tests, verify in Swagger. Then debug one existing test in VS Code. Bonus: run the gateway e2e suite and read `gateway.e2e-spec.ts`.

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20. Learning Resources

Node.js / npm

- Official docs: https://nodejs.org/docs/latest-v20.x/api/
- npm basics: https://docs.npmjs.com/getting-started

TypeScript

- Handbook (start with "TypeScript for JS Programmers"): https://www.typescriptlang.org/docs/handbook/intro.html

NestJS (the most important one)

- Official docs — read First Steps → Controllers → Providers → Modules → Guards → Interceptors → Pipes → Exception filters: https://docs.nestjs.com
- Config: https://docs.nestjs.com/techniques/configuration · Validation: https://docs.nestjs.com/techniques/validation
- OpenAPI: https://docs.nestjs.com/openapi/introduction · Passport/JWT: https://docs.nestjs.com/recipes/passport
- Testing: https://docs.nestjs.com/fundamentals/testing

Databases

- node-oracledb docs: https://node-oracledb.readthedocs.io/en/latest/
- mssql (tedious) docs: https://www.npmjs.com/package/mssql
- Oracle data dictionary (ALL_TAB_COLUMNS/ALL_ARGUMENTS) reference: https://docs.oracle.com/en/database/oracle/oracle-database/19/refrn/

Auth

- JWT introduction: https://jwt.io/introduction
- passport-jwt: https://www.passportjs.org/packages/passport-jwt/

Tooling

- Jest: https://jestjs.io/docs/getting-started · Supertest: https://www.npmjs.com/package/supertest
- Docker: https://docs.docker.com/get-started/
- VS Code Node debugging: https://code.visualstudio.com/docs/nodejs/nodejs-debugging

Video (well-maintained, beginner-friendly)

- NestJS official course hub: https://courses.nestjs.com/ (paid, by the framework authors)
- freeCodeCamp "NestJS Course for Beginners" (YouTube, free): https://www.youtube.com/watch?v=GHTA143_b-s

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21. Requires Knowledge Transfer

Confirmed from source code (you do NOT need anyone for these)

Architecture, every endpoint and its Oracle/SQL mapping, the auth journey, configuration surface, error handling, testing, local/no-DB development, Docker build — all documented above and verifiable in the repo. A large amount of *live-verified* business behavior (flexfield rules for op 65, PERSON_ID requirements for ops 36/67, date-track rules, staging DB defects) is preserved in `AGENTS.md`, DTO doc-comments, and the Postman collection's captured real responses.

Requires team knowledge — ask for these explicitly

#	Item	Why it matters	Who probably knows
1	**Production/staging infrastructure**: servers/jumpstations, how deployments are performed (no CI/CD exists in the repos), who operates Docker hosts	you cannot ship without it	previous developer / DevOps
2	**All real credentials**: Oracle, Users DB, MOTC SMS DB (password was shared via SMS), LDAP bind account, Entra client secret, production `JWT_SECRET`	app cannot connect	DevOps / DB team / client
3	**MOTC_SMS fixed INSERT values**: `MOTC_SMS_APP_ID`, `MOTC_SMS_SUBJECT_ID`, priority/language/recipient-type codes	OTP rows currently insert NULL/defaults for these; the government gateway may require exact values	client (MOTC integration owner)
4	**Cerner**: real base URL per environment + API contract confirmation	appointments module is 503 until set	client / Cerner team
5	**Corporate HTTP SMS gateway contract** (legacy OTP path)	`SmsOtpDeliveryAdapter` is a placeholder	client
6	**Oracle/SQL schema ownership + defect process**: staging defects exist in `SCHOOL_FEE_PR` (ORA-01403), `ADD_OR_UPDATE_PHONE` (rejects all phone types), `UPDATE_DEPENDENT_PR` (intermittent ORA-00027) — correct requests fail inside the DB	you'll be blamed for DB bugs otherwise	Oracle apps team
7	**Business rules not encoded anywhere**: op-51 (skipped — why?), exact role assignment rules (who gets APPROVER), MPIN client-side hashing algorithm	product correctness	client / mobile team
8	**Mobile app team contact + release coordination**: response-shape changes must be coordinated (e.g. HTTP-200-on-business-failure convention, addressType twin exception)	breaking the app silently	mobile team
9	**Security review decisions**: MPIN stored as received (legacy equality), in-memory token revocation, `trustServerCertificate` usage — accepted trade-offs that may need revisiting	compliance	previous developer / security
10	**docker-compose env gap**: newer vars (MOTC_SMS_*, OTP_STORE, DIAGNOSTICS_ENABLED, AUTH_STATIC_LOGIN) are not in the compose `environment:` pass-through — decide whether to add them or manage via `.env` on servers	deployment surprises	DevOps
11	**Git remote / branching / code review process** — the repos have no CI config; the workflow is undocumented	collaboration	previous developer

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End of handover document. Companion file: `PROJECT_ANALYSIS.md` (raw technical analysis). Keep `AGENTS.md` updated as you learn — it is the living memory of this project.